

Details		Criteria				
Type	Description	Weight	Due	Length	Use of AI	
Written exam	Final exam Final exam	50%	Formal exam period	2 hours	AI prohibited	
Outcomes assessed: LO1 LO2 LO4 LO5						
Q&A following presentation, submission or placement	Term paper presentation Students presents the findings of their research paper and answer questions about it.	30%	Multiple weeks	5-10 minutes	AI prohibited	
Outcomes assessed: LO3						
Research analysis	Term paper Summary and critical assessment of a behavioral economics article or research paper.	20%	Week 13 Due date: 31 May 2026 at 23:59	1,500 words	AI allowed	
Outcomes assessed: LO1 LO2 LO4 LO5						

Assessment summary

Term paper

Use of generative artificial intelligence (AI)

You can use generative AI tools for open assessments. Restrictions on AI use apply to secure, supervised assessments used to confirm if students have met specific learning outcomes.

Refer to the assessment table above to see if AI is allowed, ~~for assessments in this unit and check Canvas~~ for full instructions on assessment tasks and AI use.

Term paper presentation (it may be updated)

- Time: 20 minutes
- Not all group members need to present, but I can ask questions to any group member at any time.
- Structure:
- title

-outline (what the paper is about in two sentences and the structure of the presentation)

-motivation

-literature review (how does this research relates to previous research, particularly to concepts seen in class)

-methods (experimental design/empirical strategy)

-results

-interpretation of the results and conclusion
- This topic is closed for comments.

→ abstract
why they studying this → introduction
bonus points
不同点。
可逆 hypothesis
文章有图表要加
和 real problems
未来研究方向...



$$\max_{x_1, x_2, \dots, x_n} E V(x_1, x_2, \dots, x_n) \quad \text{s.t.} \quad Px = M \quad \Leftarrow \text{economic experiments}$$

"bounded rationality" 有限理性

		player 2	
		left	right
player 1	up	2, 2	0, 3
	down	3, 0	1, 1

prisoner's dilemma
pure strategy NE (down, right)

规则改变 utility also depends on others' payoffs $\Leftarrow$ "social preferences"

		player 2	
		left	right
player 1	up	\$2, \$2	\$0, \$3
	down	\$3, \$0	\$1, \$1

utility may not depend on the money one gets

Intertemporal consistency

Paradoxes related to expected utility

- Ellsberg Paradox
- Allais Paradox

} bounded rationality

Week 2

Outline

= social preferences

定义

{ differences
 samples

• social norms ...

• Reciprocity def.
 positive ~~~~~ good $\Rightarrow$ good
 negative ~~~~~

intension

• Guilt aversion def. $\Rightarrow$ universal
 /
 bad emotion People behave prosocially to avoid feeling guilty for disappointing others.

$\Rightarrow$
 need to know others' expectation

• Self image & social image $\Leftarrow$ others are looking
 $\Downarrow$
 impressing others
 show off

shallow meritocracy

- A:
 - 好学校
 - 名师辅导
 - 家庭条件好
 - B:
 - 条件一般
 - 没有额外资源
- 结果:
- A 得 90 分
 - B 得 80 分

Shallow meritocracy 的判断:
 "A did better, so A deserves more."
 ✗ 不会问:
 • 起点是否公平?
 • 学习资源是否对等?

bounded rationality
 • efficiently irrational } How the brain works
 • Dual Process theory multiple series

- ① Dual Process Theories
- ② Efficient coding hypothesis

system 1 ~~~~~ maybe for survival
 system 2 ~~~~~

limited energy $\Rightarrow$ brain work most efficient way
 not most rational way

mechanical criticism of Dual process theory



whats a valid economic theory according to Milton Friedman

According to Milton Friedman, a *valid economic theory* is **not judged by the realism of its assumptions**, but by **the accuracy of its predictions**.

does dual process theory make good predictions

DPT has limited predictive power.

whats a falsifiable theory

A falsifiable theory is a theory that makes clear predictions that could, in principle, be shown to be false by evidence.

Week 5

risk

positive : how things are $\leftrightarrow$ descriptive

normative : how things should be $\leftrightarrow$ prescriptive

prospect theory $\rightarrow$ changes on wealth (gains, losses)
(得失)

Expected utility theory $\rightarrow$ states of wealth

Week 10

market design

- matching markets
- auctions

Week 2

Fehr, Ernst, and Gary Charness. "[Social preferences: fundamental characteristics and economic consequences.](#)" *Journal of Economic Literature* 63.2 (2025): 440-514.

Week 3

Elster, Jon. "[Social norms and economic theory.](#)" In *Handbook of monetary policy*, pp. 117-133. Routledge, 2020.

Week 4

Glimcher, Paul W. "[Efficiently irrational: deciphering the riddle of human choice.](#)" *Trends in cognitive sciences* 26, no. 8 (2022): 669-687.

Grayot, James D. "[Dual process theories in behavioral economics and neuroeconomics: A critical review.](#)" *Review of Philosophy and Psychology* 11, no. 1 (2020): 105-136.

Week 5

Kahneman, Daniel. "[Maps of bounded rationality: Psychology for behavioral economics.](#)" *American economic review* 93, no. 5 (2003): 1449-1475.

Week 6

Kable, Joseph W. "[Valuation, intertemporal choice, and self-control.](#)" *Neuroeconomics* (2014): 173-192.

Week 7

Oprea, Ryan. "[Decisions under risk are decisions under complexity.](#)" *American Economic Review* 114, no. 12 (2024): 3789-3811.

Week 8

Cass R. Sunstein, [Nudging: A Very Short Guide](#), 37 J. Consumer Policy 583 (2014)

Hansen, P. G., & Jespersen, A. M. (2013). [Nudge and the manipulation of choice: A framework for the responsible use of the nudge approach to behaviour change in public policy.](#) *European journal of risk regulation*, 4(1), 3-28.

Week 9

[Maternal depression, women's empowerment, and parental investment: evidence from a randomized controlled trial, by Baranov, Balhotra, Biroli and Maselko](#)
[Will There Be Blood? Incentives and Displacement Effects in Pro-Social Behavior, by Lacetera, Macis and Slonim](#)[Links to an external site.](#)

Week 10

[Suboptimal behavior in strategy-proof mechanisms: evidence from the residency match, by Rees-Jones](#)

[Strategy proofness in experimental markets, by Guillen and Veszteg](#)

The Final and Replacement both consist of 36 multiple choice questions. Mark only the best answer.

Topics:

- social vs standard preferences
- gift exchange *in social*
- social preferences in context

区别 记定义
preferences 背景如: labor market
labor market + rules / regulations

- social norms: definition, explanation, criticism of models involving norms

efficient irrationality *w4(1)* 大脑的 energy constraints *how* → behavior bias
dual process theory: concept and criticism *w4(2)* system 1 2 分别做什么 为什么会有

- energy constraints and behavioural biases

- prospect theory

- loss aversion

- heuristics rule of thumb 比 profit max X simple rules

- time inconsistency

- hyperbolic discount

- beta-delta model

- present bias

- exponential discounting

- complexity *w7*

- nudge, types of nudges

- libertarian paternalism

- ethical considerations of nudging

- Baranov et al, maternal depression, main results

- Lacetera et al, blood donations, main results

- strategy proofness

- induce preferences

时间偏好一致 → exponential
非 exponential 比 hyperbolic + beta-delta
w6 beta-delta model 比 1/λ present bias
更慢 比 1/λ 更慢 1/λ
preference reversal

定义 *w8* types of nudges

哪些 nudge 不合伦理

w9(1) 大议和议会 (做什么, 怎么做, 主要结果)

w9(2) displacement effect

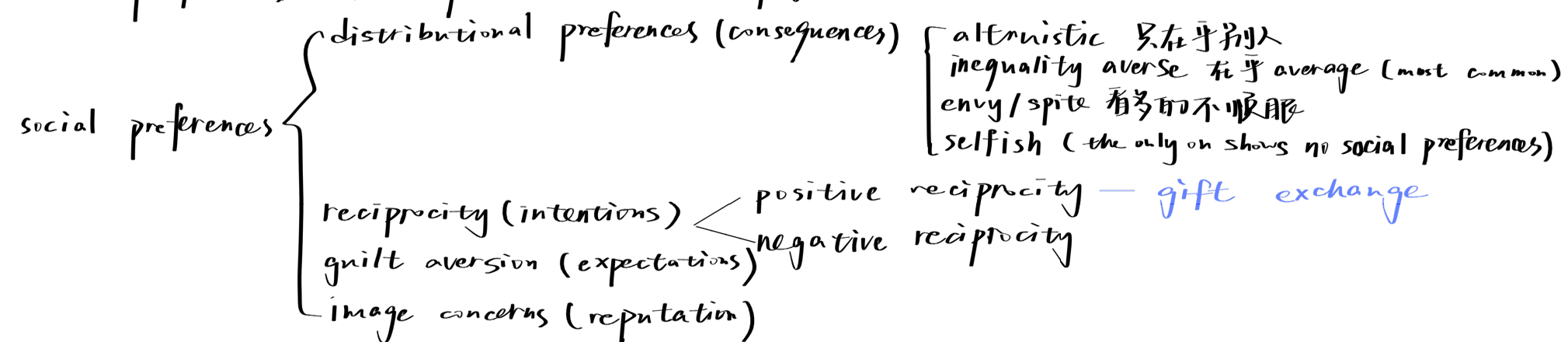
no evidence of crowding out of social preferences

domain of gains
domain of losses
reference point
different slopes
loss aversion

w10

standard preferences: self-interested (homo economicus)

social preferences: also depends on others' payoffs, actions and intentions



gift exchange: driven by (advantageous) inequality aversion

higher wages → higher efforts

从富有的公司拿钱, 而非雇主, 削弱了 gift exchange

importance of intentionality: 机器随机决定 不同于 雇主故意选择

social preferences in context:

- downward (nominal) wage rigidity
 - during recessions 裁员而非 wage cuts (negative reciprocity)
- peer comparisons
 - 不利工: 歧视 → effort ↓
 - 有利 → effort ↑ (disadvantageous inequality aversion)
- crowding out
 - explicit rules undermining trust
- legal regulations change reference points
 - set minimum wage → new fair wage

Week 3

Elster, Jon. "Social norms and economic theory." In *Handbook of monetary policy*, pp. 117-133. Routledge, 2020.

rational actions: to achieve Y, do X

social norms: do X or don't do X (non outcome oriented)

sustained by: ① external sanctions (punishment)

② internalized emotions

X moral norms (utilitarian ethics)

X legal norms (X police v members of the general community)

X private norms

criticisms of economic models involving norms

X use norms to manipulate others for personal benefit

Elster: 都认同 norms, 有约束力

X 遵守 norms to avoid punishment

Elster: ① 没人看时也遵守

② punishing is costly 都自利 → 没人去惩罚违规者

X norms exist to promote common interests (Pareto-improvements)

Elster: 许多 norms $\rightarrow$ 所有人环境 $\downarrow$

Week 4

Glimcher, Paul W. "[Efficiently irrational: deciphering the riddle of human choice.](#)" *Trends in cognitive sciences* 26, no. 8 (2022): 669-687.

brain energy constraints $\rightarrow$ behavioral biases (心理缺陷)
trade-off between the metabolic costs of computational precision and the benefits of accuracy
(optimal evolutionary)
efficiently irrational

Grayot, James D. "[Dual process theories in behavioral economics and neuroeconomics: A critical review.](#)" *Review of Philosophy and Psychology* 11, no. 1 (2020): 105-136.

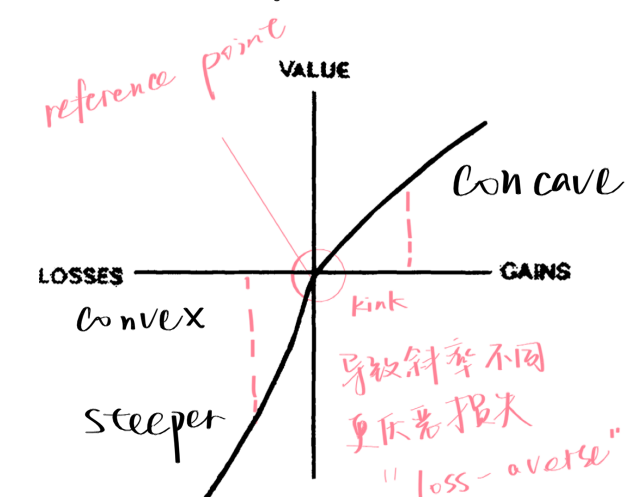
system 1: fast, reactive, automatic 情感 (behavioral biases)
system 2: slow, controlled, reflective 理性思考 (inner rational agent)

criticize: ① cross-cutting 共享神经网络
② system 2 不总理性 (过度思考, 偏向理性折扣)
③ system 1 不免非理性 (快速反应很重要)

Week 5

Kahneman, Daniel. "[Maps of bounded rationality: Psychology for behavioral economics.](#)" *American economic review* 93, no. 5 (2003): 1449-1475.

system 2 非常耗能 $\rightarrow$ 松懈 $\rightarrow$ system 1 自动省力接管 $\rightarrow$ behavior biases



prospect theory: changes of wealth $\checkmark$
states of wealth $\times$ (expected utility theory)

heuristics: rules of thumb
system 1
simple rules $\checkmark$
profit max $\times$ (rational agent)

Week 6

Kable, Joseph W. "[Valuation, intertemporal choice, and self-control.](#)" *Neuroeconomics* (2014): 173-192.

model ① exponential discounting

$$DU(x, t) = \delta^t U(x) \quad (0 \leq \delta \leq 1)$$

rational/stationarity $\Rightarrow$ time consistency

model ② hyperbolic discounting $\Leftarrow$ 实验发现人和动物 time inconsistency (X 新信息 $\vee$ 时间流逝)

$$DU(x, t) = \frac{U(x)}{1 + kt}$$

短时间延迟时贴现率极高 (极度无耐心)
(越近越心急)

preference reversal 例 $w=50$ \$100 $\times$ today \$100 $\vee$ smaller-sooner
 $w=51$ \$110 $\vee$ tomorrow \$110 $\times$ larger-later

model ③ quasi hyperbolic discounting
(beta-delta model)

$t=0$ $DU(x, 0) = U(x)$ 完全不打折, 活在此时

$$t>0 \quad DU(x, t) = \beta \delta^t U(x)$$

$\beta < 1$ 代表 present bias (对 immediate rewards 的极度偏爱, 不是立刻能拿到的心理折扣)

$t=0$ 时 $t=1$ β 发挥作用 打大折 $t=0$ 更好

$t=30$ 时 $t=31$ β 打折一样 $t=30$ 更好

Week 7

Oprea, Ryan. "[Decisions under risk are decisions under complexity.](#)" *American Economic Review* 114, no. 12 (2024): 3789-3811.

classic lottery anomalies $\rightarrow$ non-standard risk preferences

(loss aversion
four-fold pattern)

Oprea : risky x complex v

lottery treatment

100 boxes

G 90

90 boxes \$25

10 boxes \$0

→ EV (expected value) 22.5

multiple price list

(MPL)

\$15	G 90
\$16	G 90
⋮	
A	B

但实际你愿意花多少 (交换) CE (certainty equivalent)

A/B = 选一

最近得出心理值

same for mirror treatment

实验结果表明

mirror treatment 不到 22.5 就选 A 和 lottery treatment 一致

其他解释: confusion

老师解释: confusion driven by complexity

⇒ Oprea : complexity

Week 8

Cass R. Sunstein, [Nudging: A Very Short Guide](#), 37 J. Consumer Policy 583 (2014)

Hansen, P. G., & Jespersen, A. M. (2013). [Nudge and the manipulation of choice: A framework for the responsible use of the nudge approach to behaviour change in public policy](#). *European journal of risk regulation*, 4(1), 3-28.

nudge : choice architecture → alters people's behavior in a predictable way
liberty-preserving (free to choose)

	transparent (可识别意图/手段)	non-transparent (未意识到环境改变)
type 1 automatic	fly in urinal sign on floor relaxing music	wastink defaults glass shape
type 2 reflective (需要判断/思考/注意)	prompted choice save for tomorrow	

libertarian paternalism 使用 nudge, 确保可以 opt out 保障自由

ethical considerations:

X① invisible/covert (subliminal advertising/hidden triggers)

X② manipulative

X③ abuse of power (elitist politics of the brain)

Week 9

[Maternal depression, women's empowerment, and parental investment: evidence from a randomized controlled trial](#), by Baranov, Balhotra, Biroli and Maselko

maternal depression : during pregnancy or within the first year of birth

thinking healthy programme - THP

what: provide CBT in rural Pakistan

how: 16 home visits starting in the third trimester

results: ① 7 years post-intervention, depression ↓ 17%

② financial empowerment ↑

③ parental investment : time investment ↑ money investment ↑

④ heterogeneity: impacts were larger for mothers who were pregnant with girls

[Will There Be Blood? Incentives and Displacement Effects in Pro-Social Behavior](#), by Lacetera, Macis and Slonim [Links to an external site.](#)

local effect: incentive (gift cards) ↑ → donation ↑

displacement effect: 45% not new donors (they shift their donation toward incentivized sites) ⇒

! no crowding out: 不挤兑 (deferred) 并没有上升

overestimate
the total
effect on
aggregate
donations

Week 10

strategy proofness: reporting true preferences is a weakly dominant strategy regardless of what others do
└ deferred acceptance - DA
top trading cycles - TTC

inducing: 赋予学校不同货币值 → true ROL

mechanism: 运行算法 (DA/TTC)

detection: RDA/RTTC (要求说反话最优) → 还说真话说明之前没懂
(naive truth-telling)